

Windows Registry Cheat Sheet

HKCU\SOFTWARE\Microsoft\Windows\CurrentVersion\Explorer

Subkey	Description
\ComDlg32	
\LastVisitedPidlMRU	Tracks apps used to open/save files
\OpenSavePidlMRU	Tracks files accessed via open/save dialog
\MountPoints2	Tracks data about removable devices (USB, etc.)
\RecentDocs	Recent file interaction
\RunMRU	Start → Run history
\TypedPaths	Paths manually entered into Explorer address bar
\UserAssist	Per-user evidence of execution (GUI programs)
\WordWheelQuery	Explorer search history (Windows 11 22H2 / Server 2022 and earlier)

HKCU\SOFTWARE\Microsoft\Terminal Server Client\Servers

- Tracks RDP targets to which the user connected

HKCU\SOFTWARE\Classes

- %USERPROFILE%\AppData\Local\Microsoft\Windows\UsrClass.dat “plugs in” here

HKCU\SOFTWARE\Classes\Local Settings\Software\Microsoft\Windows\Shell

Subkey	Description
\MuiCache	Per-user GUI execution evidence (no execution time)
\BagMRU	Explorer folder navigation history (which folders were browsed)
\Bags	Explorer folder view settings (layout, columns, sorting, size)

Additional ShellBags subkeys in this location track the Desktop and Network Locations (UNC Paths):

HKCU\SOFTWARE\Microsoft\Windows\Shell

Subkey	Description
\BagMRU	Explorer folder navigation history (which folders were browsed)
\Bags	Explorer folder view settings (layout, columns, sorting, size)

HKCU\SOFTWARE\Microsoft\Windows\CurrentVersion\Run & RunOnce

- Starts programs on user logon (affects **CURRENT USER** only)

HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\Run & RunOnce

- Starts programs on user logon (affects **ALL USERS**)

Windows Registry Cheat Sheet

HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\Schedule\TaskCache

- \Tree = Task names mapped to GUIDs
- \Tasks = GUID-keyed metadata (actions, triggers, run times)
- Common persistence mechanism; entries can survive deletion of the on-disk task

HKLM\SYSTEM\ControlSet00x\Enum\USB

← USB VID / PID

HKLM\SYSTEM\ControlSet00x\Enum\USBSTOR

← USB Class ID / iSerialNumber

HKLM\SYSTEM\ControlSet00x\Enum\USBSTOR\Ven_Prod_Version\iSerialNumber\Properties\{83da6326-97a6-4088-9453-a1923f573b29}\####

- **0064** = First Install (Windows 7+)
 - Also found in setupapi.dev.log
- **0066** = Last Connected (Windows 8+)
 - Also \Enum\USB\VID_XXXX&PID_YYYY last write time of USB iSerialNumber key
 - Also \MountPoints2\{GUID} last write time of key
- **0067** = Last Removal (Windows 8+)

HKLM\SYSTEM\MountedDevices

- Find iSerialNumber to obtain the **Drive Letter** of the USB device
- Find iSerialNumber to obtain the **Volume GUID** of the USB device

HKLM\SOFTWARE\Microsoft\Windows Portable Devices\Devices

- Find iSerialNumber and then look for FriendlyName to obtain the **Volume Name** of the USB device

HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\EMDMgmt

- Key will ONLY be present if system drive is NOT an SSD
- Traditionally used for ReadyBoost
- Find iSerialNumber to obtain the **Volume Serial Number** of the USB device
 - The Volume Serial Number will be in decimal – convert to hex
 - You can find complete history of Volume Serial Numbers here, even if the device has been formatted multiple times
 - The USB device's iSerialNumber will appear multiple times, each with a different Volume Serial Number generated on each format

Using the **Volume GUID** found in **SYSTEM\MountedDevices**, you can find the **user** that actually mounted the USB device:

HKCU\SOFTWARE\Microsoft\Windows\CurrentVersion\Explorer\MountPoints2

While not a registry artifact, note that USB First Time Device Connected Logs are also available:

Location: **C:\Windows\inf\setupapi.dev.log**

Search for the device's **iSerialNumber** within these logs to determine the first time the device was connected.

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HKLM\SYSTEM\ControlSet00x\Control\TimeZoneInformation

HKLM\SYSTEM\ControlSet00x\Control\ComputerName\ComputerName

HKLM\SYSTEM\ControlSet00x\services\LanmanServer\Shares

- Stores all Network Shares

HKLM\SYSTEM\ControlSet00x\services\Tcpip\Parameters\Interfaces

- Stores network interface configuration information (record the interface GUID)

Network Location Awareness (NLA) has been present since Windows Vista and is used to build and manage network profiles and assist firewall profile selection. The Windows Firewall uses this information to apply firewall rules to the appropriate profile. Using NLA registry keys, you may find evidence of networks a machine has connected to, depending on retention and profile creation.

Registry key last write times may provide a rough indicator of recent activity for a network profile.

HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\NetworkList

- \Signatures
 - \Unmanaged (record **DefaultGatewayMac**, **DnsSuffix**, **FirstNetwork** (SSID), **ProfileGuid**)
 - \Managed
- \Nla
 - \Cache
- \Profiles

Network profile data is stored under **\Profiles\{GUID}** and correlated using the **ProfileGuid** value from related **NetworkList** signature entries. Per-network profile details, including **DateCreated** and **DateLastConnected**, are stored here as well.

Network type values observed include:

0x06 = Wired
0x17 = Broadband
0x47 = Wireless

On pre-1803 Windows 10 systems, legacy HomeGroup data may also be stored under:

HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\HomeGroup

Windows Registry Cheat Sheet

Prefetch is an evidence of execution artifact stored on disk, but its configuration is stored in the registry and managed by this key:

HKLM\SYSTEM\ControlSet00x\Control\Session Manager\Memory Management\PrefetchParameters

EnablePrefetcher Value:

- **0** = Disabled
- **1** = Application prefetching enabled
- **2** = Boot prefetching enabled (default on Windows Server 2003 only)
- **3** = Application and Boot prefetching enabled (default)

How Prefetch processing works:

- Task Scheduler calls Windows Disk Defragmenter every three (3) days
- When idle, lists of files and directories referenced during boot process and application startups are processed
- The processed result is stored in **Layout.ini** in the Prefetch directory, and is subsequently passed to the Disk Defragmenter, instructing it to re-order those files into sequential positions on the physical hard drive